INCH-POUND

MIL-DTL-1163C
Amendment 1
8 June 2009
SUPERSEDING
MIL-DTL-1163C
8 April 2008

DETAIL SPECIFICATION

BARS, FLINDERS

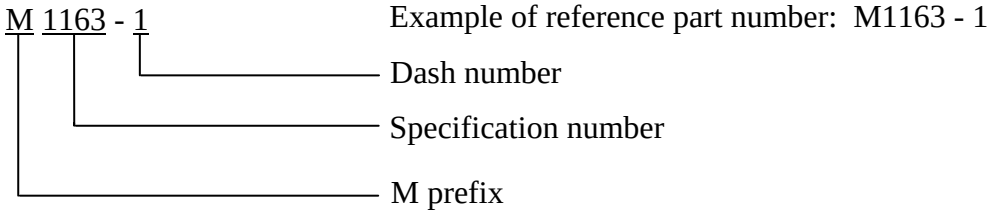
Inactive for new design after 26 December 1990

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of soft iron flinders bars. They are intended for use with compasses.

1.2 Part or identifying number (PIN). The PIN to be used for the flinders bars acquired to this specification is created as follows:



The part number had previously been JANB1163.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Government standard. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government drawings. This other government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract (see 6.2).

DEFENSE LOGISTIC AGENCY DRAWINGS

13873-03-0001 - Mercury Free.

(An electronic version of document 13873-03-001 will be made available at time of solicitation via the DLA cFolders distribution system found at the following website: https://pcf1.bsm.dla.mil/cfolders)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are that cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM A 341/A 341M	-	Standard Test Method for Direct Current Magnetic Properties of Materials using D-C Permeameters and the Ballistic Test Methods.
ASTM A 596	-	Standard Test Method for Direct-Current Magnetic Properties of Materials using the Ballistic Method and Ring Specimens.
ASTM A 773	-	Standard Test Method for dc Magnetic Properties of Materials using Ring and Permeameter Procedures with dc Electronic Hysteresigraphs.
ASTM A 848	-	Standard Specification for Low-Carbon Magnetic Iron.

(Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article testing inspection in accordance with paragraph 4.2.

3.2 Certification.

3.2.1 Material composition or magnetic property. The actual certification of material composition or magnetic property evaluation shall meet the requirements in ASTM A 848 and be included with shipment of the finished flinders bars.

3.2.2 Mercury free. Certification of mercury free material in accordance with paragraph 3.10 shall also be included with the finished flinders bars.

3.3 Material composition. Flinders bars shall be made of the highest grade commercially pure (99.6 percent) soft iron in accordance with ASTM A 848, type 1, and worked in such a manner as to avoid introduction of mechanical strain due to shear or otherwise.

3.4 Recycled, recovered, or environmentally preferable materials. Unless otherwise specified herein, all materials covered by this specification shall be new. However, recycled,

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.

recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.5 Magnetic properties. Magnetic testing shall be conducted in accordance with either ASTM A 341, ASTM A 596, ASTM A 773, or by use of a coercimeter.

3.6 Heat treatment. The bars shall be heated slowly and uniformly to a temperature between 1500 and 1600 ºF (815.6 and 874.1 °C). Cooling shall be accomplished in the furnace at a rate not to exceed 350 ºF (194.4 °C) per hour until the temperature reaches 1000 ºF (537.7 °C). The cooling rate from that point shall then proceed at a rate not to exceed 100 ºF (55.5 °C) per hour until the temperature inside the furnace reaches 400 ºF (204.4 °C). At that point, the bars shall be removed from the furnace and cooled in still air to approximately ambient temperature. The physical properties of the bars shall meet the requirements specified in ASTM A 848.

3.7 Dimensions. The completed flinders bars shall be 24 inches (609.6 mm) in length overall and shall be comprised of four sections, each of which shall be cylindrical and 2 inches (50.8 mm) in diameter throughout. One section shall be 12 inches (304.8 mm) in length, another section 6 inches (152.4 mm) in length, and two sections 3 inches (76.2 mm) in length.

3.8 Workmanship. Workmanship shall be of highest quality and each section shall have an approved black lusterless finish.

3.9 Case. Unless otherwise specified (see 6.2), the flinders bar case shall be rectangular in shape, of hardwood, well seasoned and finished; the top, sides and bottom shall be 0.5 inch (12.7 mm) thick; the sides of the case shall be dovetailed and secured together by two inside hinges approximately 2 inches (50.8 mm) from each end; the hinges shall be secured with brass screws; the whole exterior surface of the case shall be varnished; and when the case is closed, the top shall be secured to the bottom by a brass hook and screw eye near each end of the case. The case shall be approximately 26 inches in length, 5.63 inches in width and 3.25 inches deep (660.4 x 142.9 x 82.6 mm), all measurements overall. The inside of the case shall be lined with green felt and shall be divided into two longitudinal sections, into one of which shall fit the flinders bar. In the other section, there shall be four sections of smooth turned hardwood, similar in lengths to the sections of the flinders bar, but of 1.875 inches (47.6 mm) diameter. The grain of these wood spacers shall be approximately parallel to the length of the spacer. The ends of each longitudinal section of the case inside shall have fitted and secured into them a piece of gray felt about 0.5 inch (12.7 mm) thick to take up any longitudinal movement or thrust of the bar in transport.

3.10 Mercury Free. All components of the flinders bars, its case, and other included parts shall be mercury free in accordance with 13783-03-0001.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required (see 3.1), the number of units to be provided for inspection shall be specified (see 6.2). Units provided for first article inspection shall be subjected to the examinations of paragraph 4.4.1. The presence of any dimension or bar not within tolerance shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall be performed in accordance with the inspection provisions set forth herein. A sample of bars shall be selected from each lot in accordance with MIL-STD-1916 inspection level I and examined for attributes. The sample bars shall meet the requirements defined in paragraphs 3.7 and 3.8 when subjected to the examinations of paragraph 4.4.1.

4.4 Test methods.

4.4.1 Examination of bars. Each of the sample flinders bars selected in accordance with paragraph 4.3 shall be visually and dimensionally examined to verify compliance with paragraphs 3.7 and 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contact or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service’s system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The flinders bars covered by this specification are intended to for use with compasses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specific issue of individual documents referenced (see 2.2).
c. Whether a first article sample is required (see 3.1).
d. Case, if different (see 3.9).
e. Number of units required for first article inspection (see 4.2).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

compass
iron
magnetic
soft

6.4 Changes from the previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy – SH
Army – AV
DLA - GS

Preparing Activity:
DLA - GS1

(Project 6605-2009-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST data base at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:45Z
Check the source to verify that this is the current version before use.